26. Levels of Abstraction

The Concept of Level



Descriptions of complex systems can be divided into **levels**.

Upper levels are **abstractions** of lower levels.
Lower levels are **realizations** of upper levels.

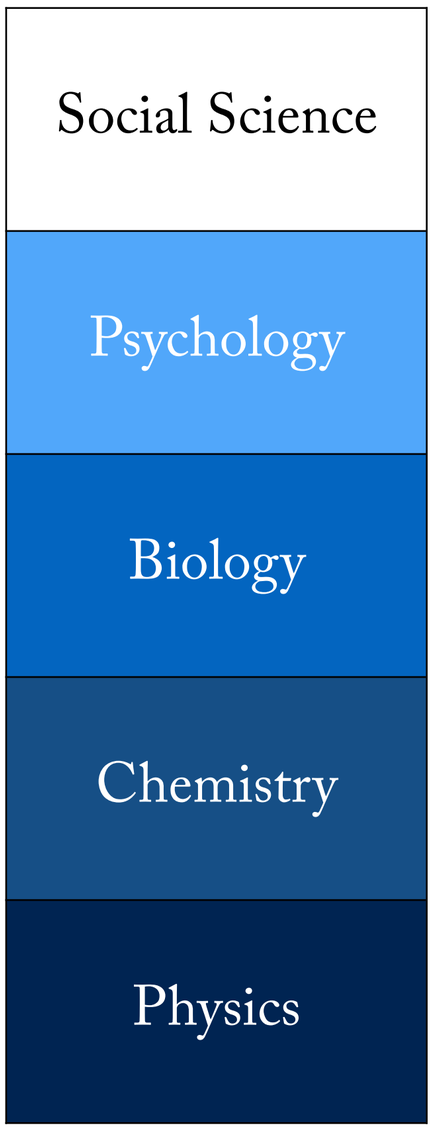
Upper levels capture **real patterns** that emerge from lower levels.

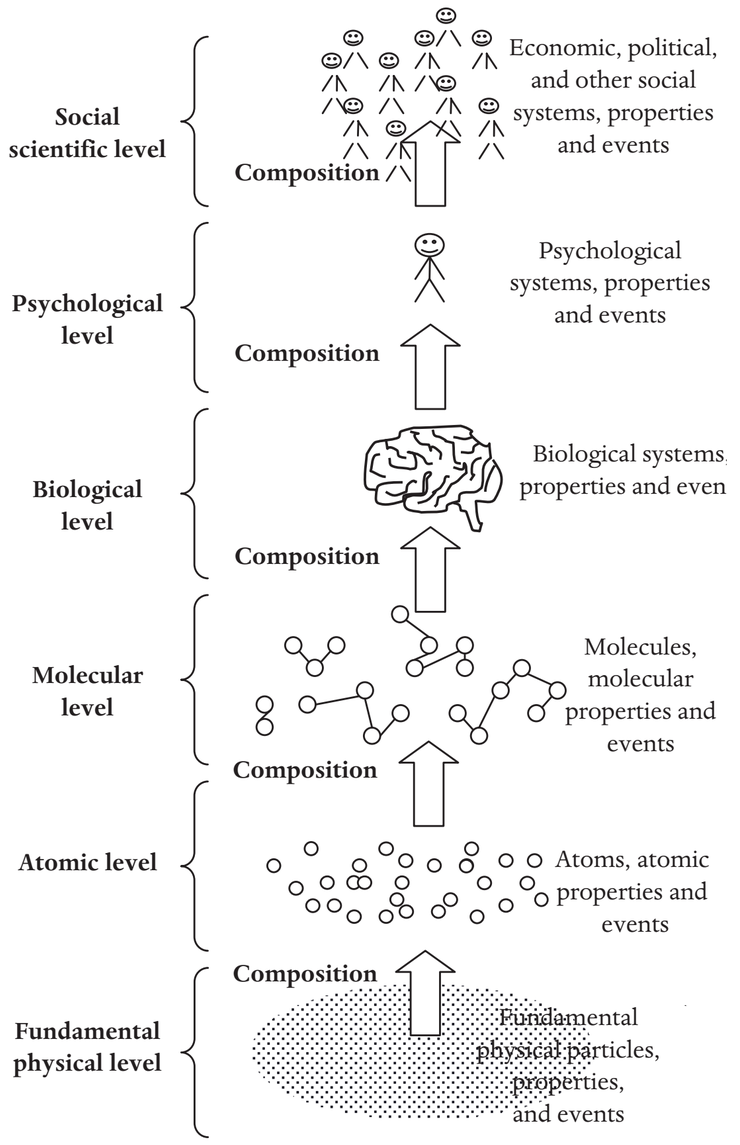
Levels in Nature

The natural sciences as a whole are organized into **levels of abstraction**, also known as **levels of explanation.**

Every level is **necessary**. A complete explanation of reality requires the contributions of all levels, not just the bottom level of physics.

Every level is **autonomous**. Levels of explanation can be studied at least in part without knowledge of the levels above and below.

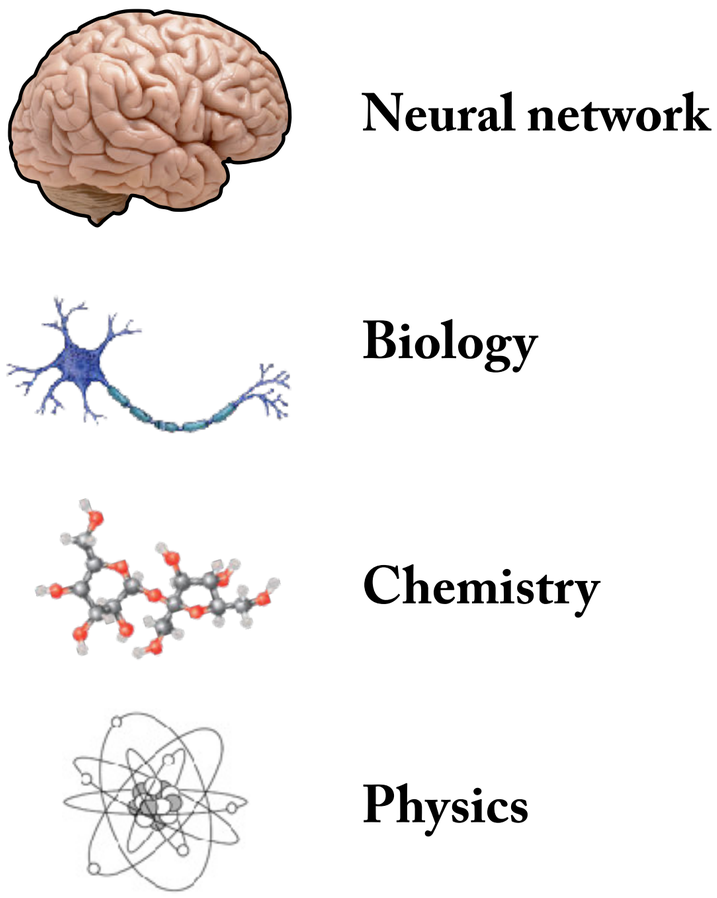




Levels in the Nervous System

The brain itself is a complex system which must be studied at many levels simultaneously.

> "If one hopes to achieve a full understanding of a system as complicated as a nervous system… then one must be prepared to contemplate different kinds of explanation at different levels of description that are linked, at least in principle, into a cohesive whole, even if linking the levels in complete detail is impractical." (Marr 1982)



Marr's Three Levels

Marr proposed an influential three-level view of cognitive science where each level aims to answer a different kind of explanatory question.

Computational theory	Representation and algorithm	Hardware implementation
What is the goal of the computation, why is it appropriate, and what is the logic of the strategy by which it can be carried out?	How can this computational theory be implemented? In particular, what is the representation for the input and output, and what is the algorithm for the transformation?	How can the representation and algorithm be realized physically?

Figure 1–4. The three levels at which any machine carrying out an information-processing task must be understood.

An Example: The Humble Cash Register

Computational Level

"The most abstract [level] is the level of what the device does and why. What it does is arithmetic, so our first task is to master the theory of addition. Addition is a mapping, usually denoted by + from pairs of numbers into single numbers."

Algorithmic Level

"In order that a process shall actually run, however, one has to realize it in some way and therefore choose a representation for the entities that the process manipulates. The second level of the analysis of a process, therefore involves choosing two things: (1) a *representation* for the input and the for the output of the process and (2) a *algorithm* by which the transformation may actually be accomplished… If the first of our levels specifies what and why, this second level specifies *how*. For addition, we much choose Arabic numerals for the representations, and for the algorithm we could follow the usual rules about adding the least significant digits first and "carrying" if the sum exceeds 9.

Hardware Level

"This brings us to the third level, that of the device in which the process is to be realized physically… The child who methodically adds two numbers from right to left, carrying a digit when necessary, may be using the same algorithm that is implemented by the wires and transistors of the cash register in the neighbor-hood supermarket, but the physical realization of the algorithm is quite different in these two cases.



Two features of levels:

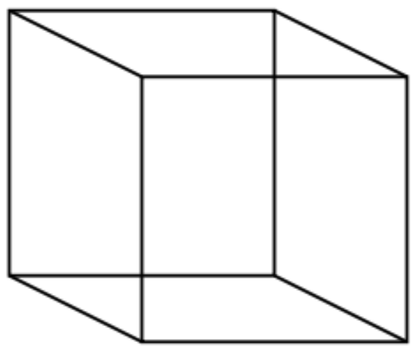
Multiple Realizability: upper levels can be realized in multiple ways at lower levels.

Methodological Independence: one can gain knowledge about different levels more or less independently.

The Cognitive-Ecological Level

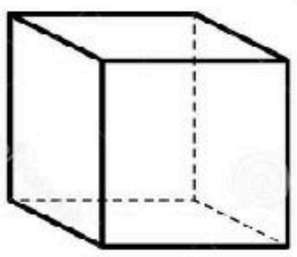
Marr called the top level of his hierarchy the "computational" level, with an eye towards *what* is computed. But this is confusing to us, since "computation" evokes algorithms and circuits. Instead, I'll call the top level the **cognitive-ecological level.** It is the level where the **cognitive task** is specified and where the **ecological rationale** for performing that task is explained.

> "It is the top level, the level of computational theory, which is critically important from an information-processing point of view. The reason for this is that the nature of the computations that underlie perception depends more upon the computational problems that have to be solved than upon the particular hardware in which their solutions are implemented. To phrase the matter another way, an algorithm is likely to be understood more readily by understanding the nature of the problem being solved than by examining the mechanism (and the hardware) in which it is embodied."

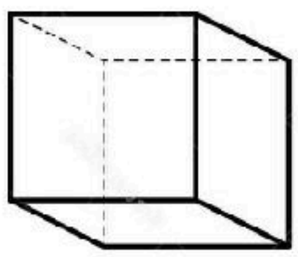


"The ambiguity of the Necker cube seems to demand a different kind of explanation. To be sure, part of the explanation of its perceptual reversal must have to do with a bistable neural network (that is, one with two distinct stable states) somewhere inside the brain, but few would feel satisfied by an account that failed to mention the existence of two different but perfectly plausible three-dimensional interpretations of this two-dimensional image."

Cognitive Level

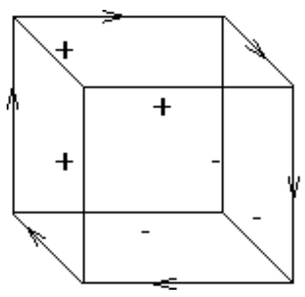


Perception 1
A cube tilted down.

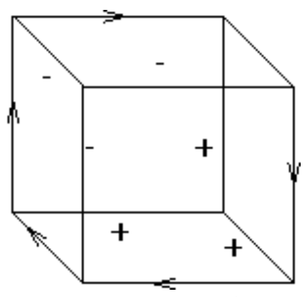


Perception 2
A cube tilted up.

Algorithmic Level



Consistent solution 1



Consistent solution 2

Physical Level



Bi-stable neural network.

- Cognition
- Computation
- Neuroscience
- Physics

The Physicalist Worldview

Marr's "computational" level is the level at which we ascribe content and rationality to an agent. Modern physicalists take this idea as the basis for their theory of the mind: cognition emerges from computation, just as biology emerges from chemistry. Mental states are entirely natural phenomena, just like cells or molecules. The natural world is a unified hierarchy of levels of abstraction.

Levels of Abstraction

Multiple Realizability

COGNITIVE-ECOLOGICAL LEVEL

"At one extreme, the top level, is the abstract computational theory of the device, in which the performance of the device is characterized as a mapping from one kind of information to another, the abstract properties of this mapping are defined precisely, and its appropriateness and adequacy for the task at hand are demonstrated."

+1

"If one hopes to achieve a full understanding of a system as complicated as a nervous system… then one must be prepared to contemplate different kinds of explanation at different levels of description that are linked, at least in principle, into a cohesive whole."

Functions

Semantic Abstraction

ALGORITHMIC LEVEL

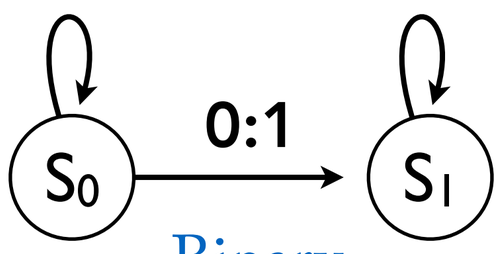
"The second level of analysis of a process, therefore, involves choosing two things: (1) a representation for the input and for the output of the process and (2) an algorithm by which the transformation may actually be accomplished."

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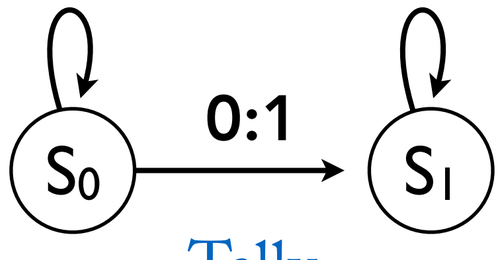
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Binary

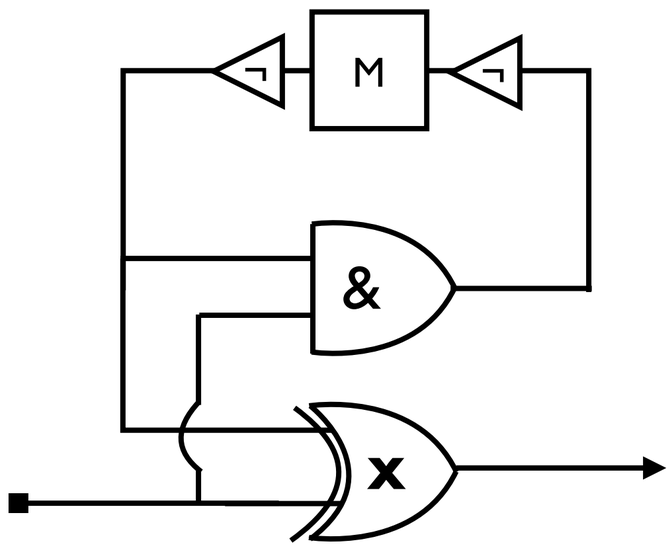


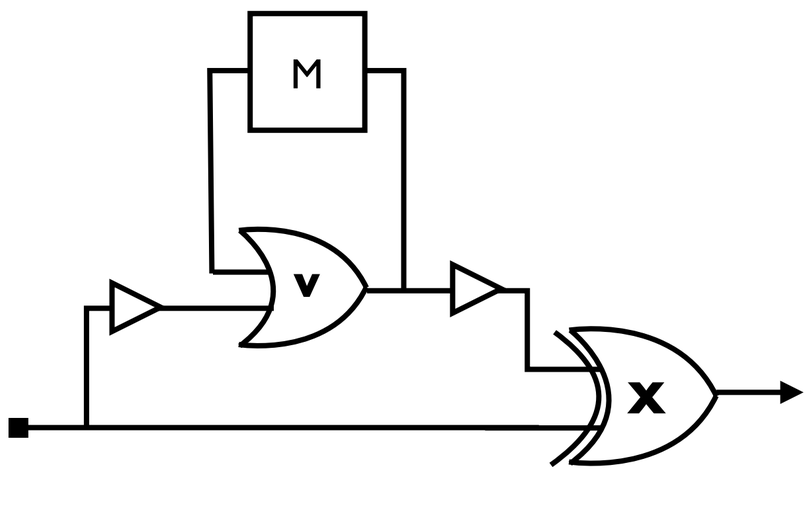
Tally

State Machines

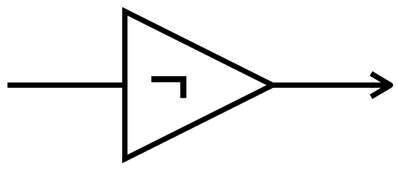
State Abstraction

MECHANICAL LEVEL





Logic Circuits

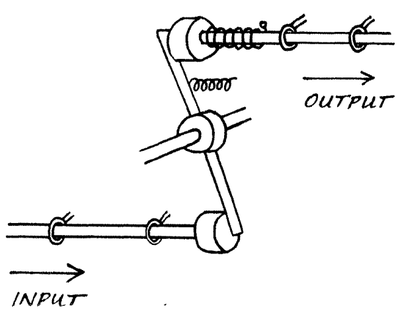


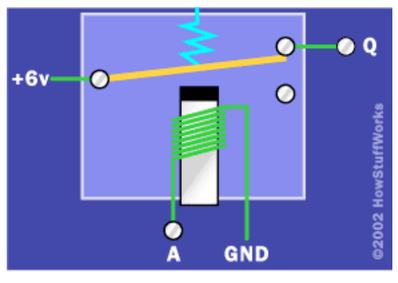
Logic Gates

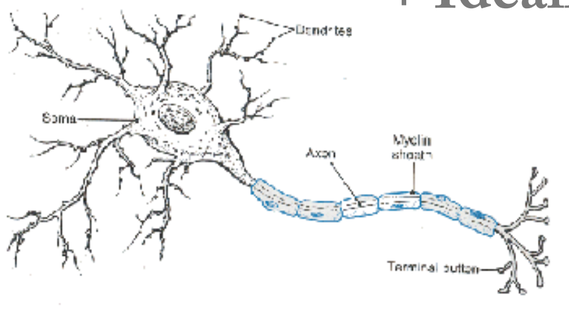
Mechanical Abstraction

+ Idealization

PHYSICAL LEVEL





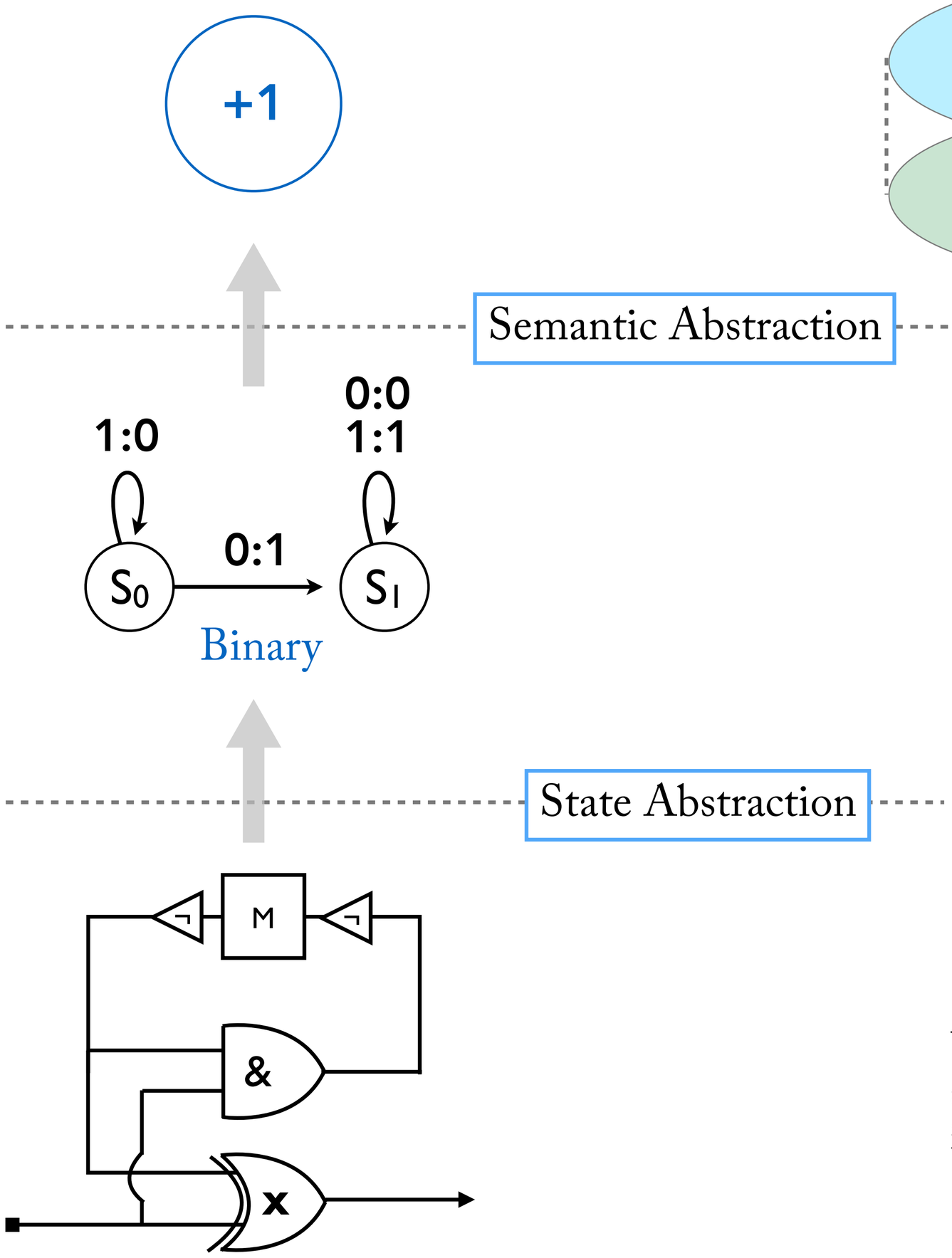


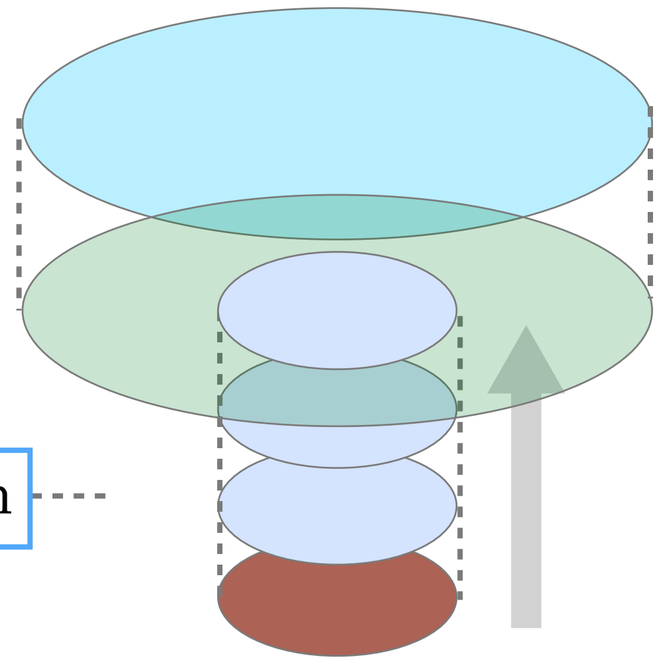
Physical Machines

"How can the representation and algorithm be realized physically?"

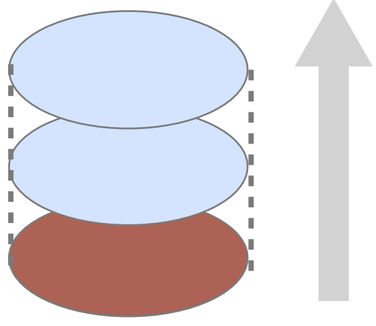
Semantic Abstraction

What are the conditions when we would say an organism computes +1? When it counts some *features of its environment*, e.g. the number of fruits in the tree or the number of steps back to the nest.





Outward abstraction abstraction over the informational relations between a system and the environment.



Upward abstraction: abstraction over a system's internal mechanisms.

The central idea here is that semantic description is just another form of functional abstraction. This brings representation and rationality into dialogue with the rest of scientific explanation.

Rational transition as abstraction. To say that a machine computes +1 is to say that, under functional abstraction, it can be characterized in terms of the +1 function. Computation is not like a metaphysical chain between the machine and the function.

Content as abstraction. The same is true for representation. Representation isn't an abstract chain (to be broken or maintained) between a symbol and its content. Content is a way of abstractly characterizing a symbol. We can describe the same physical system as a pattern of electric pulses, as the bit sequence 101, or as the number five. Each belongs to a different level of description.

In Unit 2, we've explored levels of computation: levels of abstraction and levels of computational power.

